

COFFEYVILLE, KS, USA

WMO: 724519

Lat: 37.091N

Lon: 95.566W

Elev: 229

StdP: 98.60

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 93967

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-9.2	-17.7	0.8	-10.2	-15.2	1.0	-8.1	12.1	6.6	11.3	7.1	3.5	0	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	37.6	24.2	36.0	24.4	34.0	24.2	26.4	33.2	25.7	32.6	25.2	32.0	4.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.4	20.0	30.0	23.8	19.2	29.3	23.0	18.3	28.4	83.3	33.5	80.5	32.8	77.9	32.0	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.1	9.8	8.7	-17.2	39.3	4.5	2.4	-20.5	41.0	-23.1	42.4	-25.6	43.7	-28.9	45.4		
			-17.7	27.6	4.4	1.1	-20.8	28.4	-23.4	29.1	-25.8	29.7	-29.0	30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	2.1	4.2	9.7	15.0	19.8	24.8	27.3	26.8	22.4	15.7	9.2	3.3
	DBStd	10.13	5.98	6.28	5.96	4.85	4.17	3.09	2.98	3.26	3.99	5.03	5.36	5.74
	HDD10.0	828	252	182	78	12	1	0	0	0	0	10	77	218
	HDD18.3	2193	503	397	273	120	34	1	0	1	9	113	275	466
	CDD10.0	2686	7	18	68	163	304	444	537	521	373	186	54	10
	CDD18.3	1007	0	1	5	21	79	196	279	262	132	31	2	0
	CDH23.3	10876	1	5	42	190	697	2024	3319	2997	1324	260	18	0
	CDH26.7	5084	0	1	6	36	202	869	1745	1584	576	64	2	0
Wind	WSAvg	3.9	4.1	4.3	4.8	4.9	4.0	3.8	3.3	3.1	3.2	3.6	4.1	4.0
Precipitation	PrecAvg	1063	39	40	76	111	162	148	95	81	104	85	66	55
	PrecMax	1470	108	113	187	275	286	414	207	210	233	216	189	149
	PrecMin	600	2	0	7	17	58	41	21	0	24	5	0	5
	PrecStd	192	25	32	52	57	69	83	49	45	52	55	50	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	23.5	27.2	30.1	32.5	36.3	40.0	40.4	37.6	31.1	25.8	21.0
		MCWB	12.8	15.3	17.9	19.9	22.5	25.0	23.5	23.9	23.1	22.2	17.3	15.7
	2%	DB	17.4	20.0	24.0	27.4	30.8	34.3	37.6	37.8	33.8	28.5	22.7	17.7
		MCWB	11.4	13.3	16.2	18.5	22.4	24.5	24.6	24.1	23.3	20.0	15.6	13.0
	5%	DB	14.8	17.4	22.0	25.6	28.9	32.7	36.0	36.2	32.2	26.3	20.6	14.8
		MCWB	9.2	12.1	14.6	17.8	21.7	24.3	24.7	24.1	23.0	18.6	15.2	10.6
	10%	DB	11.8	14.5	19.6	23.6	27.3	31.3	34.0	33.9	30.1	24.0	18.0	12.4
		MCWB	7.3	9.6	13.9	17.0	21.2	23.7	24.7	23.8	22.0	17.9	13.1	8.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.4	17.2	19.3	21.5	24.7	26.8	27.5	27.0	25.6	23.4	19.1	17.2
		MCDB	17.7	21.3	25.2	27.7	29.8	33.5	33.4	33.5	32.1	28.9	23.3	20.0
	2%	WB	12.5	15.1	17.7	20.3	23.6	25.9	26.5	26.0	24.7	22.0	17.3	14.4
		MCDB	16.1	18.6	22.1	25.3	28.7	32.3	33.3	33.1	31.1	26.1	20.8	17.1
	5%	WB	9.7	12.8	16.3	19.2	22.7	25.2	25.8	25.3	23.9	20.7	15.8	11.6
		MCDB	14.5	16.3	20.7	23.7	27.6	31.3	32.7	32.8	30.0	24.1	19.3	13.7
	10%	WB	7.4	9.8	14.6	18.0	21.8	24.4	25.3	24.7	23.0	19.1	14.0	8.8
		MCDB	11.2	14.5	18.8	22.4	26.3	29.9	32.3	31.8	28.6	22.8	17.6	12.1
Mean Daily Temperature Range	5% DB	MDBR	11.3	11.6	12.1	11.9	11.1	10.8	11.5	12.1	12.5	12.4	12.0	10.7
		MCDBR	16.2	15.8	15.2	14.4	12.3	12.1	13.9	14.9	14.2	14.2	14.4	14.4
		MCWBR	10.4	10.2	8.6	8.0	6.0	4.8	4.0	4.4	5.5	7.2	8.8	9.8
	5% WB	MCDBR	14.0	14.0	11.9	11.7	10.7	10.9	11.4	12.2	11.7	10.9	12.3	11.7
		MCWBR	10.1	10.1	7.8	7.6	6.0	5.1	4.4	4.6	5.1	6.6	8.7	9.6
Clear-Sky Solar Irradiance	taub	0.300	0.311	0.353	0.384	0.422	0.427	0.442	0.446	0.390	0.349	0.312	0.293	
	taud	2.471	2.452	2.373	2.314	2.268	2.277	2.282	2.245	2.367	2.464	2.506	2.508	
	Ebn at Noon	894	926	911	896	864	856	840	828	860	871	872	879	
	Edh at Noon	82	95	113	127	135	134	132	134	113	93	79	74	
All-Sky Solar Radiation	RadAvg	2.45	3.10	4.07	5.13	5.72	6.56	6.55	5.89	4.93	3.63	2.63	2.08	
	RadStd	0.23	0.48	0.24	0.34	0.53	0.58	0.42	0.38	0.38	0.44	0.30	0.23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page